COMPUTER VISION • FINAL SESSION

# How to think like a Computer Vision engineer

Exam format • sample questions • problem-solving strategy



TODAY'S GOAL

Practice the style of questions—not memorize answers.

# Your exam at a glance

Six sections • 100 marks • 120 minutes

**A • SINGLE MCQ**

## 10 marks

10 × 1 mark

**B • MULTI-SELECT**

## 20 marks

10 × 2 marks

**C • DESCRIPTIVE**

## 20 marks

10 × 2 marks • ≤100 words

**D • CODING**

## 20 marks

Choose 1 of 3 options

**E • CASE STUDY**

## 15 marks

Apply CV to a scenario

**F • NUMERICALS**

## 15 marks

Show working

**Timing tip: bank the marks you know, then return to harder questions.**

# Section A • Single-correct MCQs

10 questions × 1 mark | Choose exactly one answer

### WHAT THIS TESTS

- Core concepts and terminology
- Choosing a suitable technique
- Fast, accurate decisions

### STRATEGY

Eliminate clearly wrong choices first. Read every option before selecting.

### NEW SAMPLE QUESTION

A camera image is blurry because the camera moved during exposure. Which approach directly addresses motion blur?

A. Increase batch size

B. Apply motion deblurring

C. Use NMS

D. Convert RGB to grayscale

Discuss: why are the other options unsuitable?

**Answer: B**

# Section B • Multiple-correct MCQs

10 questions × 2 marks | Select every correct option

### HOW TO APPROACH THESE

- Judge each option independently.
- One true statement does not make the others true.
- Watch words like “always” and “never.”
- **Full marks require all correct and no incorrect options.**

### NEW SAMPLE QUESTION

Which changes can affect a CNN’s output tensor shape?

A. Input image dimensions

B. Kernel size and padding

C. Stride

D. The filename of the image

Choose all that apply, then justify each selection.

**Answer: A, B, C**

# Section C • Descriptive questions

10 questions × 2 marks | Maximum 100 words per answer

### A STRONG 2-MARK ANSWER

1. Define the concept accurately.
2. Add one distinction, reason or example.
3. Stop when the question is answered.

**AVOID**

Long introductions, unrelated details, or listing terms without explaining them.

### NEW SAMPLE QUESTION

Why might histogram equalization improve details in a low-contrast image? Mention one possible drawback.

Think about: contrast distribution, visibility, noise or over-enhancement.

**Suggested structure: “It works by… A drawback is…”**

**Practice: answer in 2–4 sentences, not a full essay.**

# Section D • Coding

20 marks | Choose ONE of three options | Hidden test cases

### THINK LIKE A DSA CANDIDATE

- Understand inputs and required output.
- Write a reusable function—not a hardcoded example.
- Handle empty input and boundary cases.
- Check indexing and zero denominators.
- Test edge cases before submitting.

### NEW PRACTICE PROBLEM

Given confidence scores, return the index of the highest score. If empty, return −1. If tied, return the first index.

**Example**

scores = [0.2, 0.8, 0.8, 0.1] → 1

**Edge cases**

[] → −1 | [0.4] → 0 | [0.5, 0.5] → 0

**In the exam: choose the option whose logic you can explain clearly.**

# Section E • Case study

15 marks | Translate a real-world need into a CV solution

### BUILD A LOGICAL CHAIN

**Problem → task → data → model → metric → failure mode**

**Ask yourself:**

- What must the system predict?
- What labels and data are available?
- Which error is most costly?
- What limits deployment?

### NEW PRACTICE SCENARIO

A supermarket wants to detect empty shelf spaces from camera images. Lighting varies, and customers sometimes block the shelves.

### DISCUSS IN PAIRS

1. Classification, detection or segmentation—and why?
2. What data variation should training include?
3. Which metric or error should be monitored?

# Section F • Numericals and problem-solving

15 marks | Show the formula, substitute values, state the answer

### A SIMPLE 4-STEP METHOD

1. Write the relevant formula.
2. Substitute the given values.
3. Calculate carefully.
4. Interpret the result or include units.

**Clear working can earn partial marks.**

### NEW SAMPLE QUESTION

A feature map is 28 × 28. A 2 × 2 max-pooling layer uses stride 2 with no padding. What is the output spatial size?

**Formula**

Output = floor((N − K) / S) + 1

**Calculation**

**floor((28 − 2) / 2) + 1 = 14**

# Let’s practise • 12-minute team challenge

Work in pairs or small groups. Explain reasoning, not just the answer.

2 min

### MCQ sprint

Pick one answer and explain why the closest distractor is wrong.

3 min

### Multi-select debate

Defend one option as true or false.

3 min

### Short answer

Write a 2-mark answer in no more than four sentences.

4 min

### Mini design review

Choose task, data, metric and one failure mode for the shelf scenario.

**Takeaway: read precisely • show reasoning • test edge cases • connect models to real tasks.**